Extra Bite #11: Inductors, Transformers, and Piezoelectric Devices

Core materials, toroids vs solenoids, saturation, eddy currents, ferrite beads, and crystals.

Group E6D · 11 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Core Materials And Why They Matter

A coil's inductance depends on the geometry of the winding and the **permeability** of the core (E6D06). Air has permeability of 1. Iron and ferrite cores have permeabilities in the hundreds or thousands, multiplying the inductance for the same number of turns. The trade-off: real cores have losses, saturate at high current, and drift with temperature.

Core type	Behavior	Notable use
Powdered iron	Best **temperature stability** of magnetic characteristics (E6D08). Iron particles bound in insulating binder; lower permeability than ferrite but very stable.	HF inductors where drift matters.
Ferrite	Higher permeability than powdered iron, so **fewer turns** for a given inductance (E6D05). Temperature stability is worse than powdered iron.	RF chokes, transformers, VHF/UHF parts.
Brass	**Decreases** inductance when inserted into a coil (E6D11). Brass acts as a shorted turn, opposing the coil's field. Used as tuning slugs for fine adjustment.	Tuning slug in adjustable coils.

Toroidal vs Solenoidal Cores

Wind the same wire on a straight rod (solenoid) and you get a coil whose magnetic field bulges out into surrounding space. Wind it on a donut-shaped core (toroid) and the field is mostly **trapped inside the core itself**.

Primary advantage of a toroidal core: the magnetic field stays inside the core (E6D10). Less stray coupling to nearby components, less RF leakage, easier to mount near sensitive circuits.

Eddy Currents And Lamination

Solid metal cores let circulating currents flow in the core itself when the magnetic field changes. Those **eddy currents** burn power as heat and lower efficiency. The fix:

Why are inductor and transformer cores sometimes built from thin layers? To reduce power loss from eddy currents in the core. (E6D04) Thin laminations insulated from each other break up the long current paths that would otherwise circulate.

Powdered-iron cores accomplish the same thing differently: individual iron particles are insulated by their binder, so eddy currents cannot link across the core.

Saturation

Push enough current through a coil and the core's iron domains all line up. Beyond that point, the core no longer contributes to inductance, the inductance collapses, and the coil looks more like a piece of wire.

What causes inductor saturation? Operation at excessive magnetic flux. (E6D12) Saturation depends on the core material and the ampere-turns. Symptom: current spikes, distortion, sometimes a hot core.

Ferrite Beads For Parasitic Suppression

A **ferrite bead** slipped over a transistor lead adds lossy impedance only at high frequencies, killing parasitic oscillation without disturbing the signal at the intended operating frequency.

Common VHF/UHF parasitic suppressor at the input and output of a transistor HF amplifier: **ferrite beads** (E6D09).

Piezoelectric Devices

Question	Answer	Q
What is piezoelectricity?	**A characteristic of materials that generate a voltage when stressed and that flex when a voltage is applied.** (E6D01) Two-way effect: squeeze it, get voltage; apply voltage, get motion.	E6D01
Aspect of the piezoelectric effect	**Mechanical deformation of material due to the application of a voltage.** (E6D03)	E6D03
Equivalent circuit of a quartz crystal	**Series RLC in parallel with a shunt C** representing electrode and stray capacitance. (E6D02) The series resonance comes from the crystal's mechanical resonance; the shunt C represents the crystal's electrode capacitance plus stray package capacitance.	E6D02

Mental Cheat Sheet

- Core permeability sets inductance.
- Powdered iron → best temperature stability.
- Ferrite → higher permeability, fewer turns.
- Brass slug → **decreases** inductance (shorted-turn effect).
- Toroid advantage: field stays inside the core.
- Eddy currents → power loss → cured by thin laminations or powdered iron.
- Saturation = excessive magnetic flux. Inductance collapses.
- Ferrite beads → VHF/UHF parasitic suppressors on transistor leads.
- Piezoelectric: voltage when stressed, motion when voltage applied.
- Quartz crystal equivalent = series RLC with shunt C across it.

Common Traps

- Powdered iron has **better** temperature stability than ferrite, not worse.
- Ferrite needs **fewer** turns than powdered iron for the same L (higher permeability).
- Brass **decreases** inductance (eddy currents oppose the field). Ferrite would **increase** inductance (high permeability). Ceramic is essentially inert – non-magnetic and non-conductive. Aluminum, like brass, would also decrease inductance.
- Saturation is caused by **flux**, not by frequency or permittivity.
- Laminations exist to fight **eddy currents**, not to "simplify assembly" or "reduce cost."
- Crystal equivalent circuit is **series RLC** (mechanical resonance), with the shunt C being just the holder.
- Piezoelectric effect is **mechanical**; do not confuse with photoelectric or photovoltaic.

Takeaway: Permeability sets inductance; powdered iron is stable, ferrite is high-permeability, brass slugs decrease L. Toroids confine the field. Eddy currents are tamed by laminations or particle insulation; saturation is caused by too much flux. Ferrite beads are the standard VHF/UHF parasitic killer. A quartz crystal looks electrically like a series RLC with a shunt C.

Bite 11 Question Bank (E6D)

E6D01. What is piezoelectricity?
A. The ability of materials to generate electromagnetic waves of a certain frequency when voltage is applied.
B. A characteristic of materials that have an index of refraction which depends on the polarization of the electromagnetic wave passing through it.
C. A characteristic of materials that generate a voltage when stressed and that flex when a voltage is applied.
D. The ability of materials to generate voltage when an electromagnetic wave of a certain frequency is applied.

E6D02. What is the equivalent circuit of a quartz crystal?
A. Series RLC in parallel with a shunt C representing electrode and stray capacitance.
B. Parallel RLC, where C is the parallel combination of resonance capacitance of the crystal and electrode and stray capacitance.
C. Series RLC, where C is the parallel combination of resonance capacitance of the crystal and electrode and stray capacitance.
D. Parallel RLC, where C is the series combination of resonance capacitance of the crystal and electrode and stray capacitance.

E6D03. Which of the following is an aspect of the piezoelectric effect?
A. Mechanical deformation of material due to the application of a voltage.
B. Mechanical deformation of material due to the application of a magnetic field.
C. Generation of electrical energy in the presence of light.
D. Increased conductivity in the presence of light.

E6D04. Why are cores of inductors and transformers sometimes constructed of thin layers?
A. To simplify assembly during manufacturing.
B. To reduce power loss from eddy currents in the core.
C. To increase the cutoff frequency by reducing capacitance.
D. To save cost by reducing the amount of magnetic material.

E6D05. How do ferrite and powdered iron compare for use in an inductor core?
A. Ferrite cores generally have lower initial permeability.
B. Ferrite cores generally have better temperature stability.
C. Ferrite cores generally require fewer turns to produce a given inductance value.
D. Ferrite cores are easier to use with surface-mount technology.

E6D06. What core material property determines the inductance of an inductor?
A. Permittivity.
B. Resistance.
C. Reactivity.
D. Permeability.

E6D08. Which of the following materials has the highest temperature stability of its magnetic characteristics?
A. Brass.
B. Powdered iron.
C. Ferrite.
D. Aluminum.

E6D09. What devices are commonly used as VHF and UHF parasitic suppressors at the input and output terminals of a transistor HF amplifier?
A. Electrolytic capacitors.
B. Butterworth filters.
C. Ferrite beads.
D. Steel-core toroids.

E6D10. What is a primary advantage of using a toroidal core instead of a solenoidal core in an inductor?
A. Toroidal cores confine most of the magnetic field within the core material.
B. Toroidal cores make it easier to couple the magnetic energy into other components.
C. Toroidal cores exhibit greater hysteresis.
D. Toroidal cores have lower Q characteristics.

E6D11. Which type of core material decreases inductance when inserted into a coil?
A. Ceramic.
B. Brass.
C. Ferrite.
D. Aluminum.

E6D12. What causes inductor saturation?
A. Operation at too high a frequency.
B. Selecting a core with low permeability.
C. Operation at excessive magnetic flux.
D. Selecting a core with excessive permittivity.

Answer Key

E6D01: C	E6D02: A	E6D03: A	E6D04: B
E6D05: C	E6D06: D	E6D08: B	E6D09: C
E6D10: A	E6D11: B	E6D12: C	